

3.1 INTRODUCTION:

The organisms have to perform various tasks in order to live. To achieve their tasks, there must be coordination and interaction among different parts or organs of the body. This results in proper activity or response.

Have you ever have touched some hot object accidentally? Can you recall what you did then suddenly? You must have removed your hand immediately from that hot object. Do you know how and why you did that? The environmental conditions are always subject to change without any warning. In order to live, the living organisms have to adjust themselves according to the changing conditions. Such adjustment would only be possible when they can detect changes, analyze them and issue commands accordingly to various organs of the body. This will make not only their survival possible but also ensure the continuity of their species.

So, coordination is defined as the process where different units of a system work together to perform a meaningful function.

Stimulus:

It refers to any factor which cause change either in internal or external environment of the organism. The changes are detected by special cells or organs termed as **receptors**. For example, eyes are photoreceptors (sensitive to light), ears are sound receptors, nose is chemoreceptor for gases and tongue is chemoreceptor for solids or liquids.

A stimulus is:

A change in the environment that can be detected by a sense organ and bring about a response

Stimulus = Heat

Response = Let go



Fig. 3.1 Stimulus

Response:

It is the activity performed by some living organism after analyzing stimulus or stimuli. It is exhibited by organs like muscles, glands which are termed as **effectors**.

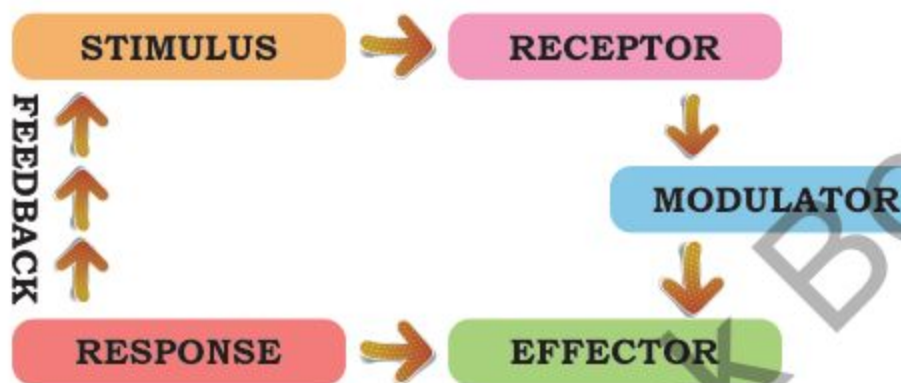


Fig. 3.2 Response activity

3.2 TYPES OF COORDINATION:

There are two types of coordination: 1) Chemical Coordination, 2) Nervous Coordination

Both are inter-related and interact with each other to develop proper response.

1) Chemical Coordination: It takes place by releasing specific, signaling molecules such as **hormones** by special cells or glands. These molecules are released to stimulate or inhibit other cells or tissues of the body. The target cells must have receptor molecules (special receiving molecules) for signaling molecules.

This kind of coordination is helpful in lower animals as well as plants because of their simple body plans and small size.

2) Nervous Coordination: It is an advance type of coordination exhibited by most of the animals, which is a consequence of specially designed cell, **Neurons**. The neurons upon stimulation generate electrochemical signals. Since the signaling is in electric form so it is very rapid.

A comparison of Nervous coordination with Chemical coordination:

NERVOUS COORDINATION	CHEMICAL COORDINATION
1) Activity of Neurons 2) Signal type is electro-chemical 3) Rapid in action 4) Response is shorter duration 5) Advance type of coordination 6) Exclusively related to animals 7) Involves neurotransmitters	1) Activity of special secretory cells 2) Signal type is purely chemical 3) Slower in action 4) Response is longer duration 5) Primitive type of coordination 6) Related to all organisms 7) Involves other signaling molecules such as hormones.

Coordination in lower organisms and plants:

Lower organisms like prokaryotes, protozoa, algae, fungi and plants have chemical coordination through signaling molecules released usually in the form of hormones to regulate their movements, growth, metabolism, reproduction, etc. For example, leaves of the “touch me not” are closed when touched. After a short while, they restore their original open position. The pressure or touch serves as stimulus while closing down of leaves is the response.

Give at least one reason why leaves of “touch me not” fold or closed when touched?



Fig. 3.3 Touch me not plant

Other responses like growth, reproduction are due to the secretion of hormones by secretory cells and carried via the transport media to their respective target organs. Another fascinating movement is the turning towards the direction of sun in “sun flower” plants. It's a bit slower movement which is believed to be caused by phytohormones called Auxins. It causes rapid cell growth in shaded regions of the plant.



Fig. 3.4 Sun flower

3.3 HUMAN NERVOUS SYSTEM

Human nervous system like other vertebrates is “centralized-type nervous system” (CNS). It is the most complicated type. Stimuli from various organs of the body are sent simultaneously its control center or central nervous system where they are integrated, analyzed and processed to develop command in the form of response. Centralized nervous system consists of two major divisions, viz. Central Nervous System (CNS) and Peripheral Nervous System (PNS).

3.3.1 Central Nervous System:

It is the major command and control center to which stimuli are reported and decisions are made and conveyed to effector organs. It consists of two main components, brain and spinal cord.

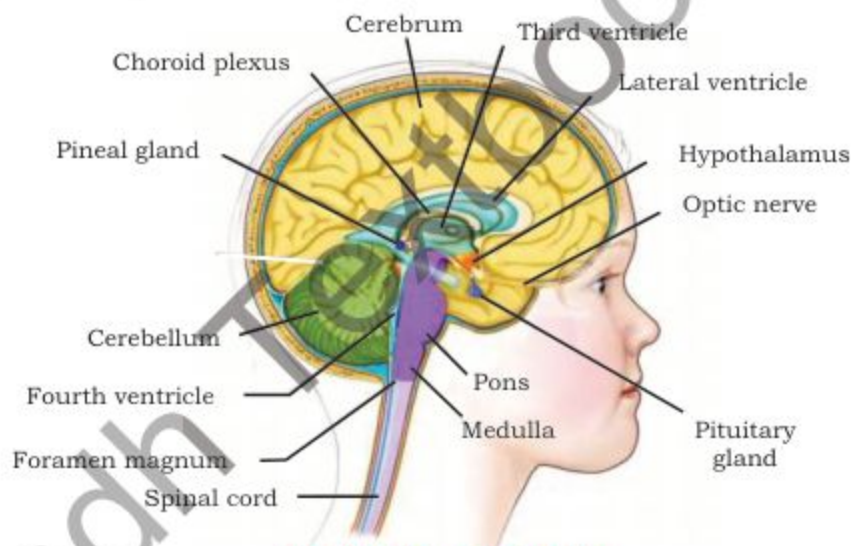
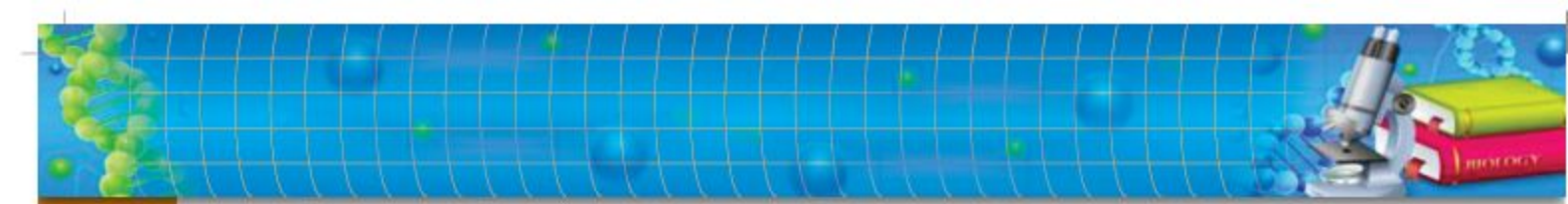


Fig. 3.5 Human brain

1. Brain: It is the major command and control center of our body. It is wrapped in three protective membranes called meninges. Inside the brain, there are empty cavities or ventricles filled with a **cerebro spinal fluid** (CSF). It also provides a cushion-like protection to the brain. Bones of the skull provide another protection to brain.

Human brain consists of following important parts: Cerebrum, hippocampus, amygdala, thalamus, hypothalamus, mid-brain, cerebellum, and medulla oblongata.



Cerebrum: It is the largest part of the brain where important decisions are made. It is considered to be the seat of intelligence, all conscious activities and memory. Its outer part, **cortex** or **gray matter** is grayish in color and consists of cell bodies of neurons while inner part **white matter** is whitish and consists of cell processes which are hair-like outgrowths. Cerebrum consists of two hemispheres, i.e. Right and left cerebral hemispheres. The right cerebral hemisphere regulates the left side of the body while the left cerebral hemisphere to the right side of the body. The cortex is associated with thoughts, plans, actions and determination. It can be divided into four sections or lobes, viz., Frontal lobe, Parietal lobe, Temporal lobe and Occipital lobe. Frontal lobe is associated with thoughts, emotions, etc. Parietal lobe is associated with different sensations like pressure, temperature, language processing, etc. The temporal lobe is involved in hearing and speech. The occipital lobe is associated with vision.

Thalamus: Thalamus lies inside in the brain just above the hypothalamus. It guides the stimuli towards appropriate part of the cortex.

Hypothalamus: Hypothalamus regulates life maintaining functions like blood pressure, body temperature, hunger, thirst, etc. It plays vital role in maintaining homeostasis of the body.

Hippocampus: It is related with long-term memory.

Amygdala: It's a deep seated small area involved in emotions (pain, pleasure, etc.)

Mid brain: In human it is relatively smaller and involved in integration of visual and olfactory (smell) stimuli. It is also collaborator of spinal cord with fore-brain.

Cerebellum: It is highly convoluted structure located on the dorsal side just below the cerebrum. It controls the precision in movement of the muscles for balance and maintains the position of the body in relation to gravity. Activities like writing, drawing, painting, dancing, crafting have become possible due to its elaborate structure in human.

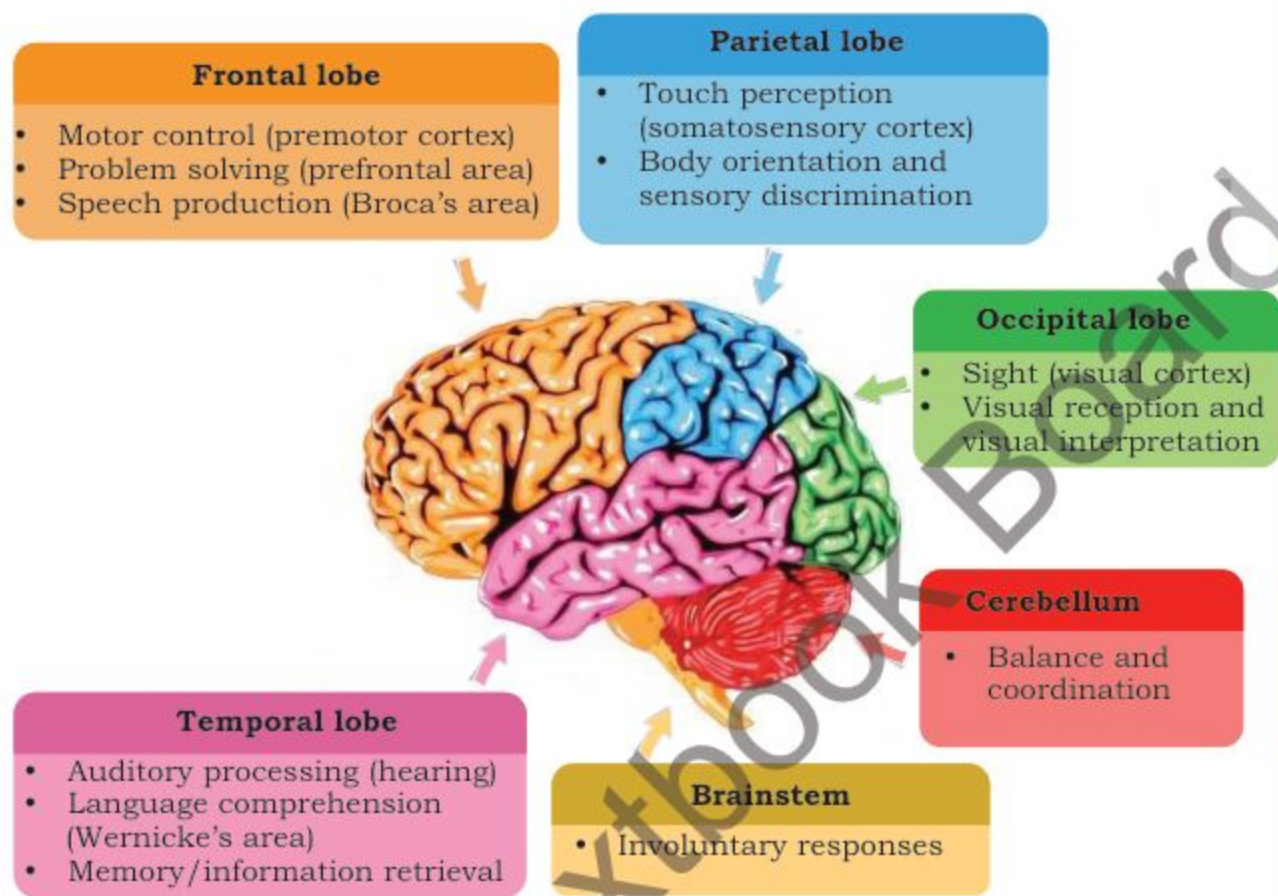


Fig. 3.6 Parts of brain

Medulla oblongata: It lies just above the spinal cord. It is the control center for automatic activities like breathing, heart-beat, blood pressure, coughing, swallowing, hiccupping, digesting food, etc. Such activities are termed as Reflexes.

Pons: It lies on the ventral side of medulla oblongata. It helps in controlling the facial muscles as well as helps in sleep and wakening.

2. SPINAL CORD: It's a butterfly shaped, thick, whitish, long tube like structure which arises from medulla oblongata and extends down through the vertebral column. Unlike brain, in cross section, its outer portion is whitish called **white matter** while inner is grayish or **gray matter**. Like brain, it is also wrapped in meninges and bathed in cerebrospinal fluid. It acts as a mini control center for few reflexes. It also

acts as express way for flow of information from brain to the different parts of the body and vice versa.

Peripheral nervous system:

It consists of cables which arise from the central nervous system and connect it to different organs of the body. Each cable is termed as a **Nerve**. Each nerve consists of bundles of axons of both sensory and motor neurons. The PNS consists of somatic nervous system and autonomic nervous systems. The somatic nervous system is associated with skeletal muscles and glands while the autonomic nervous system is associated with involuntary functions like digestion, breathing, etc. These functions are vital for maintaining life processes.

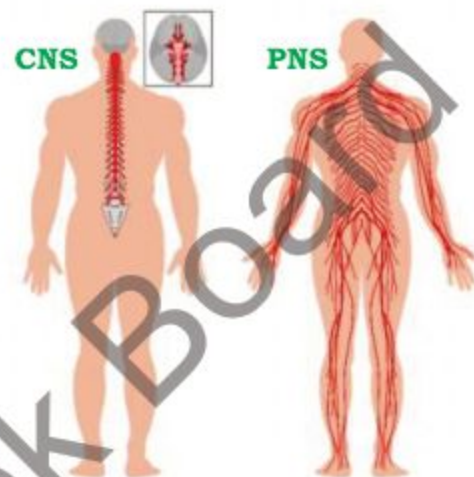


Fig. 3.7 Nervous system

3.3.2 Neuron:

The cells of the nervous system are termed as neurons. Each neuron is specialized to generate and conduct neuronal signal or **nerve impulse**. Typically, a neuron consists of a cell body or **soma** and **cell processes**. Cell body or soma comprised of plasma membrane, cytoplasm and nucleus. The cell processes are hair-like projections given out from soma. They are of two types, dendrites and axon.

Dendrites receive stimulus while axon transmits the command either to some other neuron or some **effector cells**. Like an electric cable, cell processes usually have insulated coverings, termed as **myelin sheath** to ensure the uninterrupted transmission of nerve impulse.

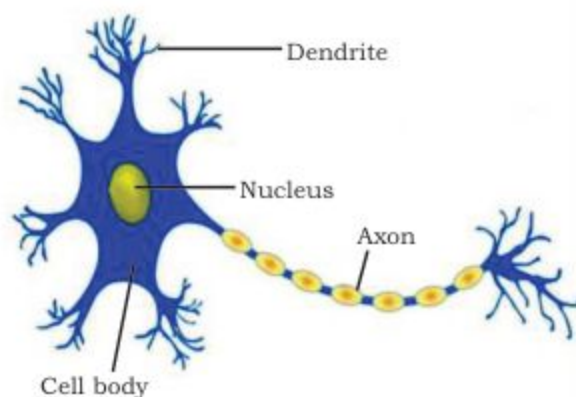


Fig. 3.8 Neuron

3.3.3 Reflex action:

You know that activities like blinking of eyes, hand withdrawal, knee-jerk, sneezing, hiccup, cough, etc are automatic. They are not learnt during the life time. Such automatic, pre-programmed responses regulated by CNS are termed as reflex actions. Some of them are directly regulated by brain while others by spinal cord. The shortest path of a reflex action, such as knee-jerk

involves two neurons, a sensory neuron and a motor neuron. The pathway of a reflex action is termed as reflex arc. In complicated reflexes, the reflex arc may involve one or numerous inter-neurons in between sensory and motor neuron.

EXPERIMENT: Contraction of shin muscle (leg muscle) of Frog using battery.

Observation: Muscles contract when provided signal through neurons by nervous system. In this experiment, we observe that a shin muscle removed from dissected frog is placed in methylene blue solution in a petri dish. When artificially stimulated by power supplied i.e. 12 volts D.C. battery, it contracts.

Apparatus: Dissecting box, Frog, dissecting tray, petri-dish, 12 volt D.C. charged battery, wires.

Procedure: Dissect out a pithed frog with the help of a teacher to expose its shin muscles as shown in diagram. Remove the shin muscle with sciatic nerve carefully and place in stretched condition in petri-dish. Connect the battery with wires and then touch the shin muscle at its

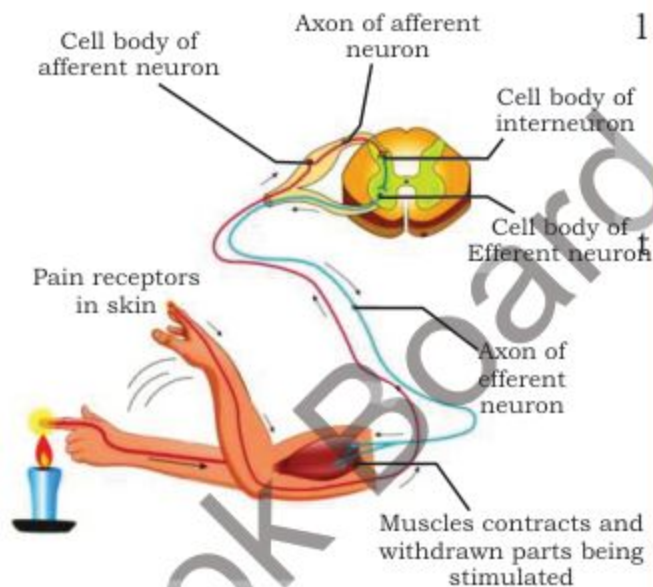


Fig. 3.9 Reflex action

beginning and end with wires. Now observe the muscle contraction. Repeat the experiment for three times.

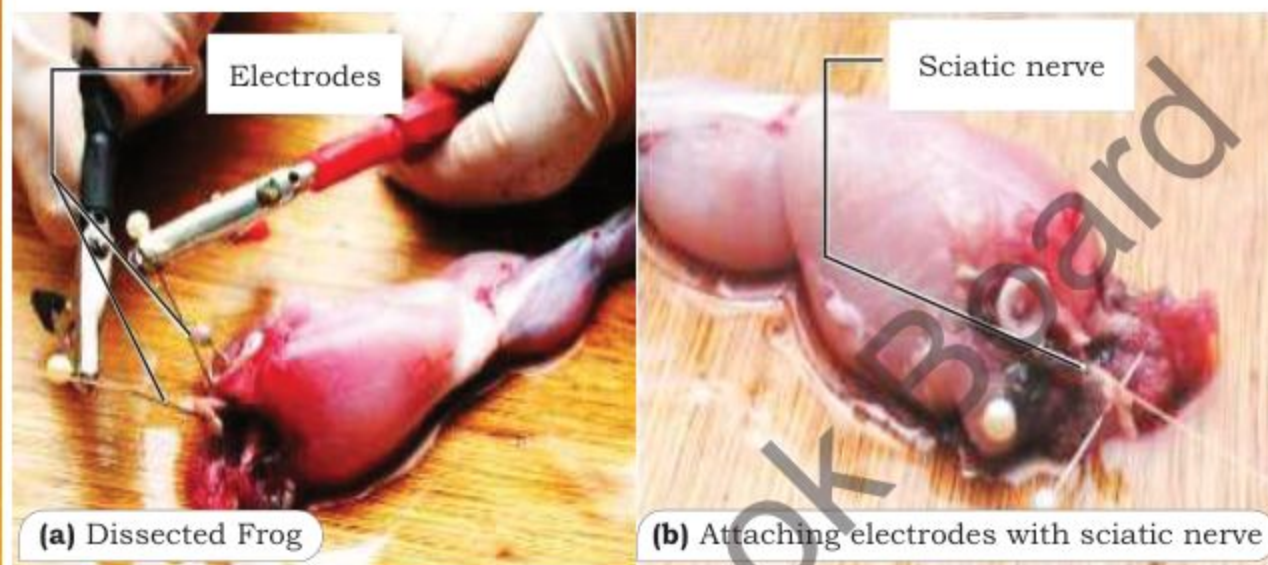


Fig. 3.10 Hind limb of frog showing sciatic nerve after dissection

3.4 RECEPTORS OF THE HUMAN BODY

Human body can detect number of stimuli like light, sound, gravity, damage to the tissues, etc, through its various receptors. We will discuss here only photoreceptor and sound receptors, i.e., eyes and ear, respectively.

Eye:

It is the organ of sight. It works on the principle of a simple camera which collects light reflected from any object in front and diverts it to a layer of sensory cells or **retina**. The light captured by retina is converted into nerve impulses and reported to the brain. Each eye lies in a bony socket for protection. A tough outer coat, **sclera** (white part of eye) covers each eye. In front, the sclera becomes transparent termed as **cornea**. Behind it, lies a small chamber, **aqueous humor** filled with watery fluid. At its back lies coloured part or **iris** with a central hole, called **pupil**. The pupil appears blackish. Immediately behind the iris is a crystalline convex **lens** suspended by a ring of circular muscles known as **ciliary body**. The

contraction of ciliary muscles causes changes in the shape of the lens to adjust focus. Behind the lens, there is a main cavity of eye ball filled with a clear gel, **vitreous humor**. The innermost layer of eye is retina on which the image is formed by cornea and lens. It has sensory cells, **rods and cones** which upon stimulation convert light signals into nerve impulses and report them to the brain.

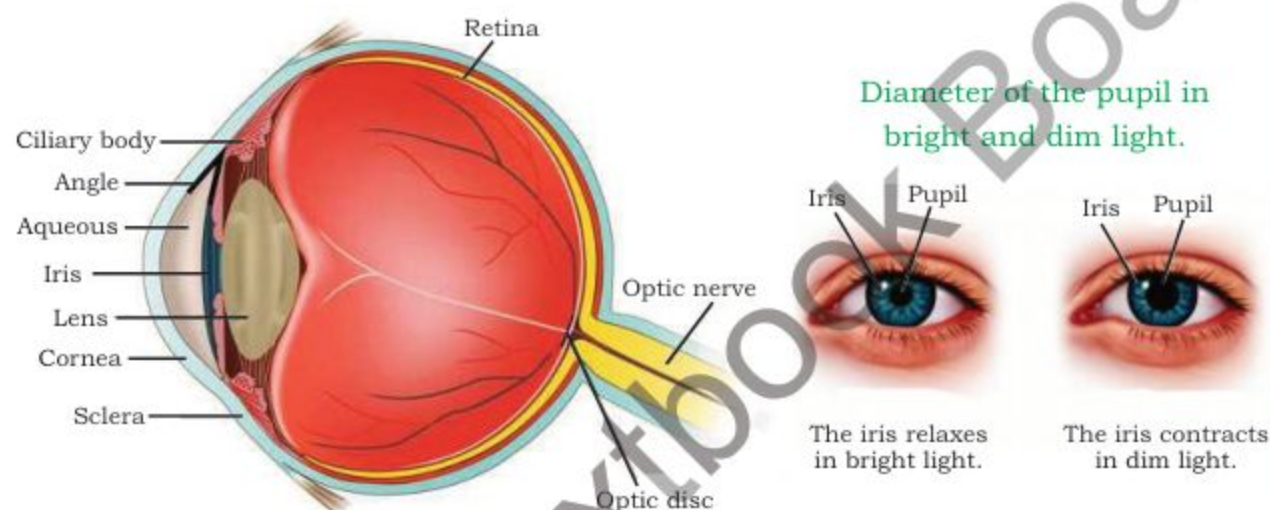


Fig. 3.11 Human eye

Pupil reflex:

The pupil adjusts itself depending upon the intensity of light. In case of bright light, it protects the retina by constricting itself so less amount of light falls on retina. Vice versa, in dim light condition, the pupil dilates to allow more light to fall on retina.

Accommodation:

It is an automatic process of altering focus to get sharper image of the near objects. To do this, the ciliary muscles contract allowing the elastic lens to become thicker and more convex. With age, the lens loses its elasticity and as a result, accommodation becomes increasingly difficult.

Role of vitamin A with vision:

Vitamin A or retinal is required for proper vision and needed for

sensory cells of retina. It also helps the cornea to be well lubricated. Deficiency of vitamin A could lead to corneal ulcers and blindness.

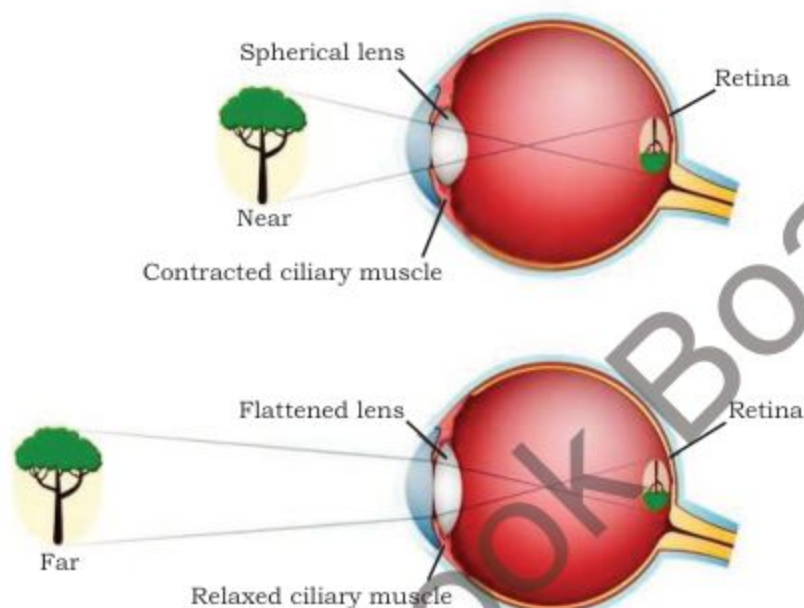


Fig. 3.12 Mechanism of accommodation

Short-sightedness and long-sightedness:

Short-sightedness (**myopia**) refers to the difficulty in focusing distant object while the near objects are focused normally. On the other hand long-sightedness (**Hyperopia**) is the difficulty in focusing closer objects while distant vision is clear. Both can be diagnosed and corrected by using appropriate glasses or contact lenses.

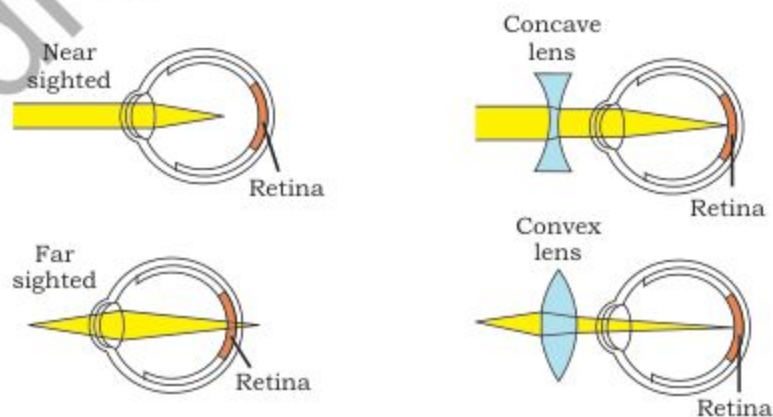


Fig. 3.13 Short sightness and long sightness

Colour blindness:

It's a deficiency of vision in which one cannot distinguish certain colours such as blue and yellow or red and green. It is due to the defect in cones of retina. Though, the vision may be normal in a colour blind person, yet the most common problem is driving because of their inability to distinguish red and green traffic lights.



Fig. 3.14 Colour blindness

Contribution of Ibn-al-Haitham and Ali-Ibn-Isa about the structure of eye and treatment of ophthalmic diseases:

Ibn-al-Haitham:

He was a great Muslim mathematician, philosopher, astronomer and physicist of 11th century. He was considered as “**father of modern optics**” due to his great contribution in principles of optics and visual perception. He was the first person to consider vision as a result of bouncing back of light from an object and then enters our eyes. His most important book on optics was “**Kitab-ul-manazir**”.



Fig. 3.15
Ibn-al-Haitham

Ali Ibn-Isa:

He was one of the most important Muslim ophthalmologists of medieval times. In his famous book “**Memorandum of the oculists**” on ophthalmology, he described more than hundred different eye diseases and their treatment.

Ear:

Ear is an organ of hearing and balance. It consists of three parts: 1) outer ear, 2) middle ear and 3) inner ear. The outer ear consists of pinna, ear canal and tympanic membrane or ear drum. The pinna composed of folds of skin and cartilage. The **pinna** leads into the ear canal which is

closed at the inner end by tympanic membrane. Ear canal has hair and produces wax to trap dust and small foreign bodies. The outer ear collects and transmits sound waves.

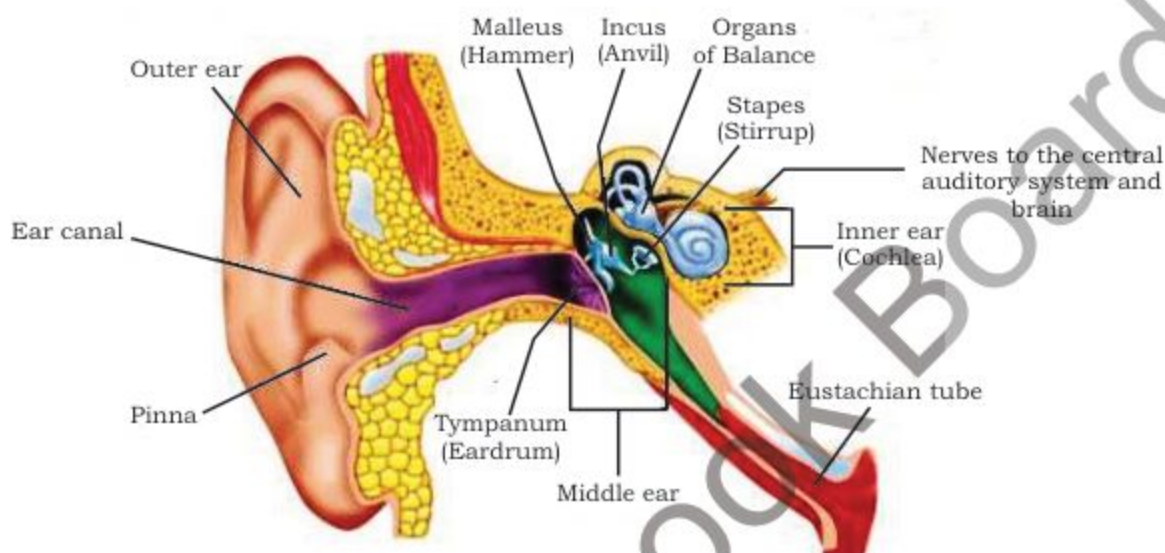


Fig. 3.16 Human ear

The middle ear consists of a small cavity containing three small moveable bones (**malleus, incus and stapes**). The middle ear is connected to inner nasal cavity through a small tube, the **Eustachian tube**. The middle ear receives sound waves from air outside and transmits it into the fluid in the inner ear.

The inner ear consists of a front membranous **cochlea** and a rear, three **semicircular canals** deep inside the skull bones. The cochlea is associated with hearing while semicircular canals are associated with balance. Both cochlea and semicircular canals are fluid filled and contain sensory hair cells. These cells transform sound waves into nerve impulse.

Role of ear in balance:

Semi circular canals are sensitive to gravity, position and movements of head. Any of such changes are detected and reported to the brain through nerve fibers. The three semicircular canals are interconnected and lie right angle to each other. They are connected to a

swollen part, the **vestibule**. Semicircular canals and vestibule are involved in maintaining balance of the body in relation to gravity.

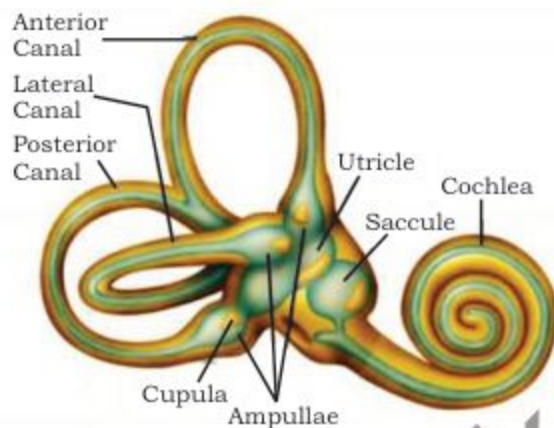


Fig. 3.17 Inner ear

Light and Sound in a thunderstorm:

In a thunderstorm, the lighting and its acoustic effect on ground appear with interval. Since the light travels much faster than sound, the lighting appears first and the sound of thunder is heard after a short while.



Fig. 3.18 Thunderstrome

3.5 ENDOCRINE SYSTEM:

It is an important means of chemical coordination. In animals, it is a system of ductless glands which secrete hormones directly into the blood. It carries hormones to their target tissues or organs. The hormones usually required in small quantity. They act like chemical signals or chemical messengers for target organs either stimulating or inhibiting their function.

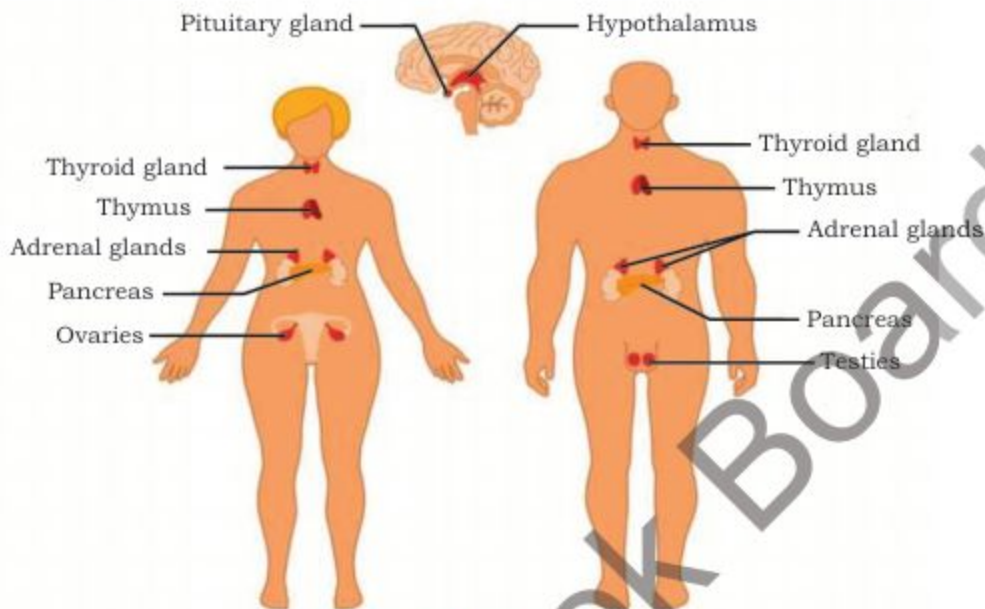


Fig. 3.19 Endocrine system

Following are important endocrine glands in human body, Pituitary gland, Thyroid gland, Pancreas, Adrenal gland and Gonads. They are located in different locations in our body.

1) **Pituitary gland:**

It is located in brain and considered to be very important. It secretes number of hormones which influence upon other endocrine glands also besides other organs. Pituitary gland consists of two lobes in human, an anterior lobe or anterior pituitary gland and a posterior lobe or posterior pituitary gland.

Anterior pituitary gland:

It has number of hormone secreting cells. Its important hormones and their effect with target organs are summarized in the table:

Hormones	Target organ	Important effects
Follicle Stimulating Hormone (FSH)	Gonads (Testes and ovaries)	Stimulate gonads to develop gametes

Luteinizing Hormone (LH)	Gonads	Development and release of gametes.
Thyroid Stimulating Hormone (TSH)	Thyroid gland	Stimulate thyroid gland
Somatotropin (Growth Hormone=GH)	Bones, Cartilages, muscles, etc	Growth in children and normal body structure and metabolism in adults
Adenocorticotropin Hormone (ACTH)	Adrenal cortex	Stimulates adrenal cortex
Melanocyte Stimulating Hormone (MSH)	Skin	Stimulates pigmentation (melanin) in skin

Some important hormones of Anterior Pituitary gland, their target organs and important effects

Posterior Pituitary gland:

It actually stores and releases some hormones of hypothalamus. Few neurons of hypothalamus store and secrete their hormones from posterior pituitary gland. Examples of such hormones are **antidiuretic Hormone (ADH)** and **Oxytocin**. ADH maintains the blood pressure, blood volume and tissue water. Oxytocin stimulates greater contraction of smooth muscles as well as social behavior.

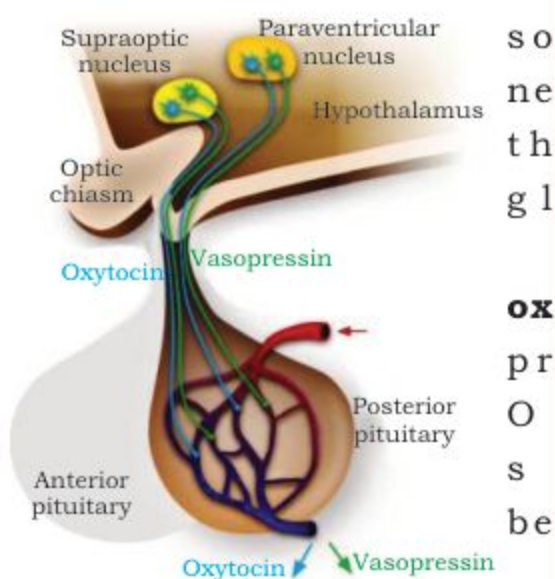


Fig. 12.20 Posterior Pituitary gland

Thyroid Gland:

It's a butterfly shaped gland located on trachea in the base of neck. It secretes **thyroxine** and **calcitonin**. Thyroxine has iodine as its important constituent. It regulates the rate of the metabolic activities of cells. It regulates the physical growth and mental development in children. In case of its deficiency, physical and mental retardation occur in children. If the intake of iodine in diet is low in adult, the thyroid gradually enlarges in size. This abnormal condition is termed as "**goiter**". Calcitonin released in response to high level of calcium in blood lowers the blood calcium.

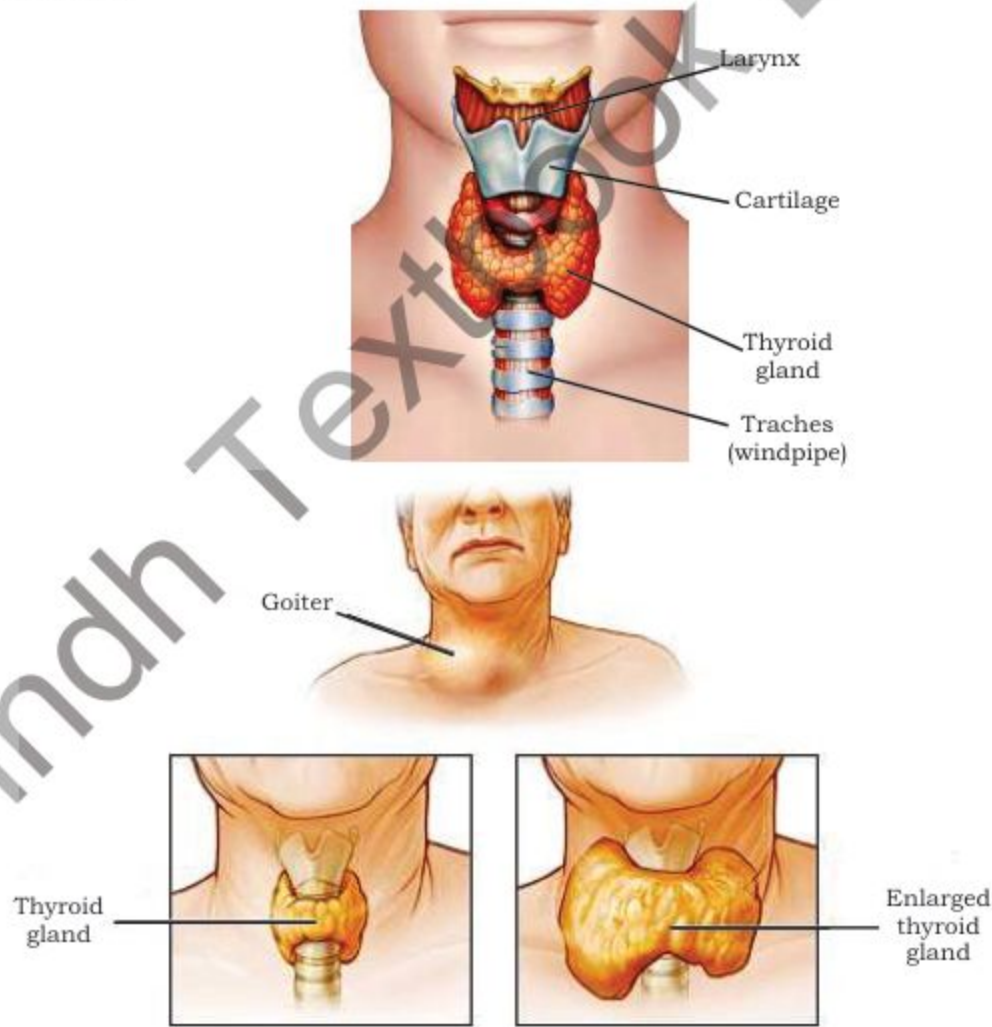


Fig. 3.21 Thyroid gland

Pancreas:

Pancreas is about 6 inches long, leaf-like in structure located in the abdominal cavity in between stomach and small intestine. It is both exocrine as well as endocrine gland in nature. The endocrine part consists of patches of cells called "Islets of Langerhan's". It is involved in regulating glucose metabolism. In response to high level of glucose, it secretes **Insulin** which helps in decreasing the blood glucose levels. On the other hand, low level of blood glucose, it secretes **glucagon** which increases the glucose level up to normal. The regulation of blood glucose through insulin and glucagon is a type of negative feed-back in which opposite effect is observed in relation to stimulus.

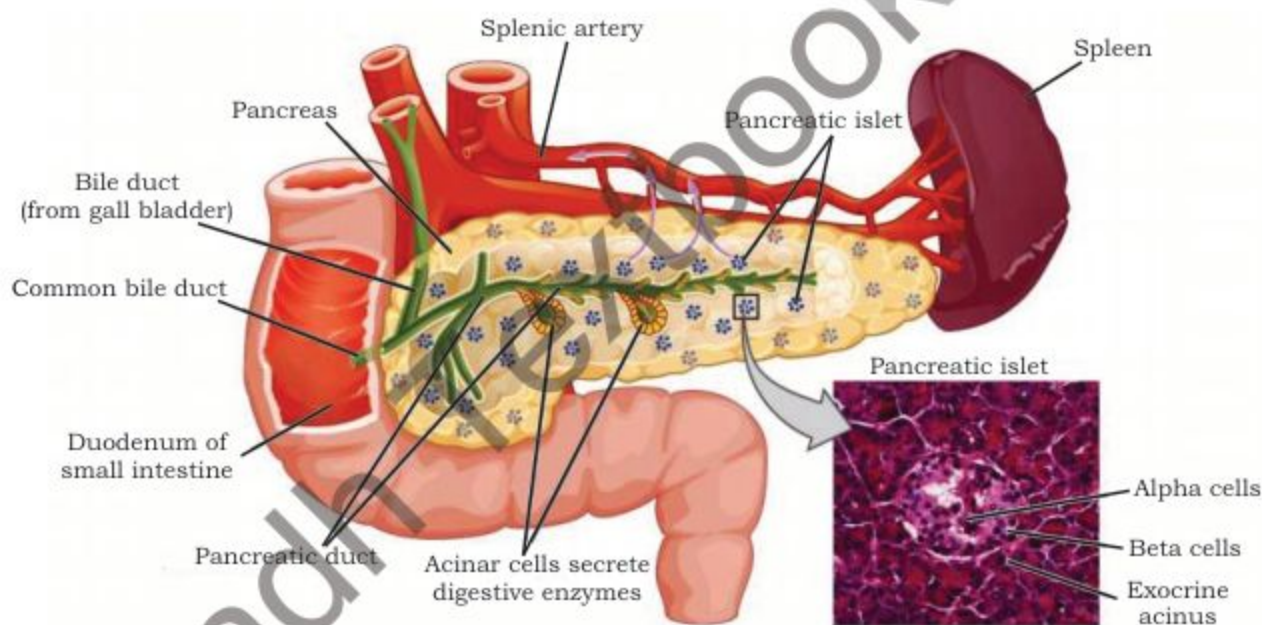


Fig. 12.22 Pancreas

Diabetes mellitus:

It's a disorder in which pancreas produces insufficient or no insulin. As a result, the level of glucose in blood becomes very high. The excess amount of glucose is excreted in urine so the frequency of passing urine increases many times. Moreover, the patient feels very thirsty and hungry. In the absence of using or storing glucose by the cells, the patient loses weight and gets tired.

WAYS TO MANAGE DIABETES MELLITUS

- **Balanced Diet**
- **Exercise**
- **Medication**
- **Blood Glucose Monitoring**

Dietary Management of Diabetes

- Dietary Management and appropriate amount of physical activity play important roles in diabetic control.
- Balanced diet with a variety of foods.
- “3 Low, 1 High” principle, i.e low fat, low sodium (or salt), low sugars and high fibre.
- Reduce intake of saturated fat, trans fat and sodium can lower the risk of developing heart diseases and hypertension.
- Controlling intake of energy for the purpose of weight reduction or maintenance.

Fig. 3.23 Dietary management of diabetes

In case of low blood insulin, it can be administered through injecting insulin derived from animals. Some patients may show allergic reactions to this animal-insulin. It has been overcome by using pure human insulin made by genetically altered bacteria.

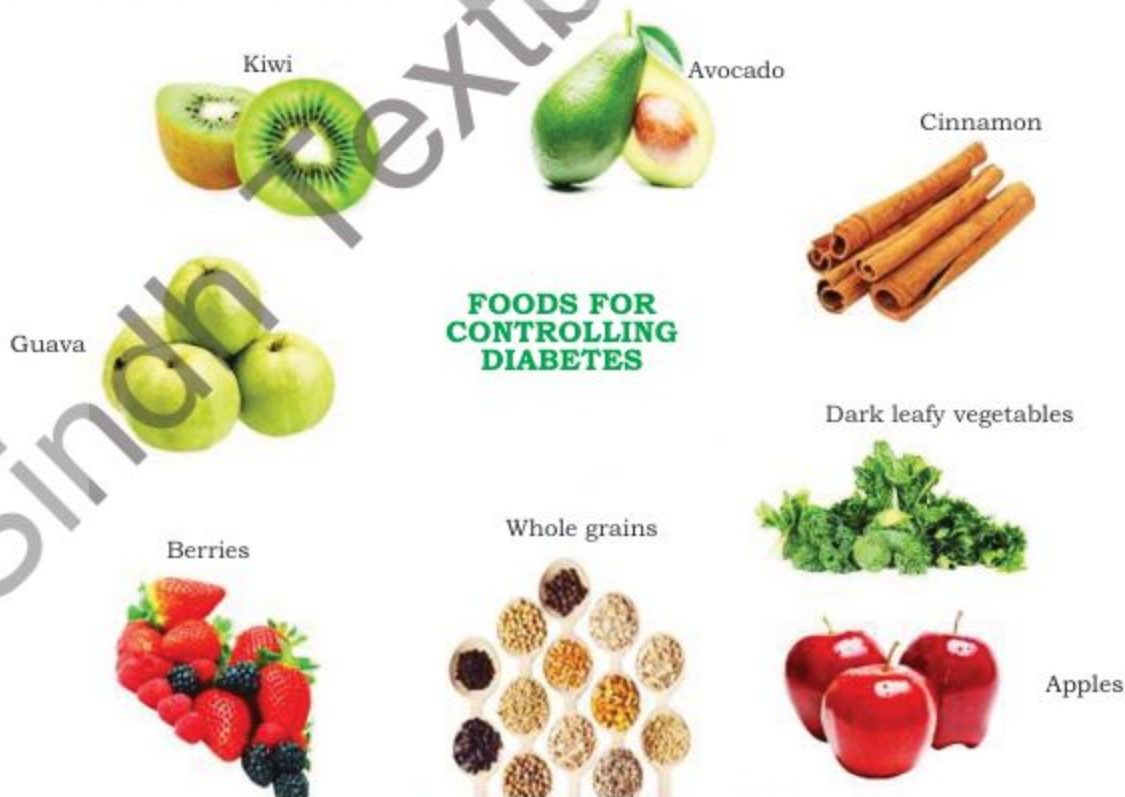


Fig. 3.24 Food for controlling diabetes

Mg/DL	Fasting	After Eating	2-3 hours After Eating
Normal	80-100	170-200	120-140
Impaired Glucose	101-125	190-230	140-160
Diabetic	126+	220-300	200+

Fig. 12.25 Blood glucose chart

ACTIVITY

Write a paper on the changes in body while performing an exercise like running a 100 meter sprint race.



Adrenal gland:

Adrenal gland is located on the top of each kidney. It is triangular shape and consists of two glands; the outer part is called **Adrenal Cortex** while the inner one is **Adrenal Medulla**. Adrenal Cortex secretes number of **steroid** (lipid) hormones necessary for regulation of body metabolism, maintenance of water and salts. **Cortisol** is one of the important hormones secreted by adrenal cortex in response to illness or inflammation of any tissue. It stimulates glucose production. Adrenal medulla responds to emergency conditions to produce so called “fight or flight response”. It secretes **adrenaline** or **emergency hormone** resulting in an increase in blood glucose level by breaking down of glycogen. As a result, the body responds to the emergency situation quickly. It dilates the

blood vessels of muscles, heart and brain and constricts those of kidneys and skin so that more blood is supplied to the muscles, heart and brain.

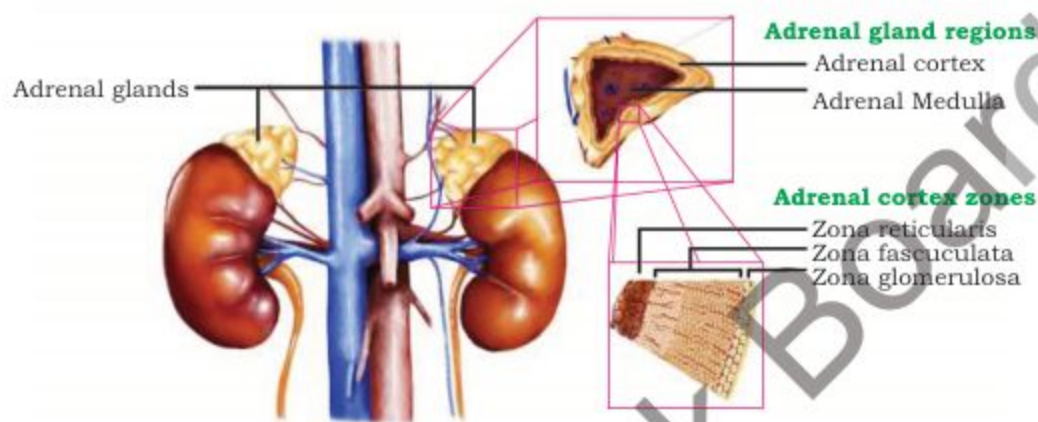


Fig. 3.26 Adrenal glands

Gonads:

Gonads are the reproductive organs, viz., testes in male while ovaries in female. They are involved in gametes formation and hormones secretion.

Testis:

Each testis is oval shaped structure located in a pouch of skin, **scrotum**. It secretes hormone known as **testosterone**, a type of androgen hormone. Testosterone is responsible for the development of secondary characteristics in boys like appearance of moustache and beard, deepening of voice, etc.

Ovaries:

Each ovary is about the size of a grape located in the lower abdominal cavity on either side of uterus. Ovary secretes **estrogen** and **progesterone** hormones. Estrogen is responsible for the development of secondary characteristics in girls like development of breasts, sharpening of voice, etc. Progesterone maintains and prepares uterus for pregnancy.

Feed back control mechanism of hormonal secretion:

Hormone secretion is regulated through feed-back control so that they are secreted whenever required. There are two types of feed-back systems working in the body, negative and positive feed-back controls.



Negative feed-back control:

It refers to the opposite effect in relation to the stimulus. For example, if there is an increase in blood glucose level (effect), the pancreas would secrete insulin (response) which will bring down the blood glucose to its set point.

Positive feed back control:

It refers to enhancement of the effect in relation to stimulus. For example, when an infant sucks the nipple of his mother's breast, she secretes hormone to secrete milk. Further continuous sucking increases the hormonal secretion several folds.

3.6 DISORDERS OF THE NERVOUS SYSTEM:

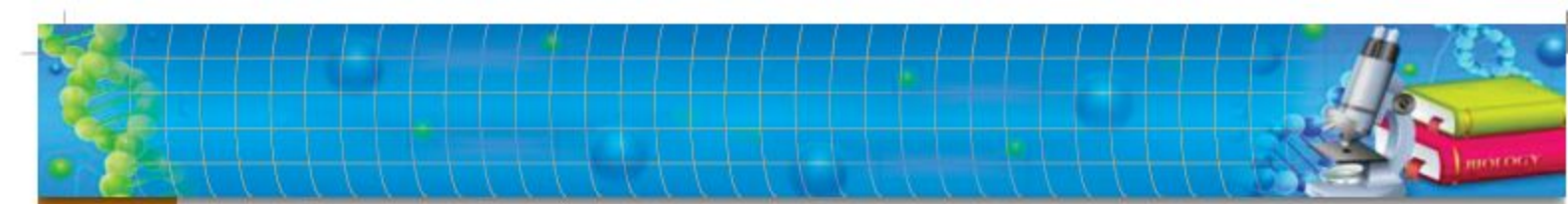
Like any other system of the body, nervous system could suffer from disorders. Depending upon the nature of disorders, they could be of two common types: Vascular (e.g. paralysis) and Functional (e.g., epilepsy).

Paralysis:

It is characterized by partial or complete loss of controlled movement caused by the inability to contract one or more muscles. It may be accompanied by loss of sensation. The most common cause is either bleeding (hemorrhage) or blood clot in the specific part of brain. However, it could be a result of injury to brain, spinal cord or nerves. It can be treated if diagnosed the exact cause. Physiotherapy of the affected muscles is required otherwise muscles may degenerate.

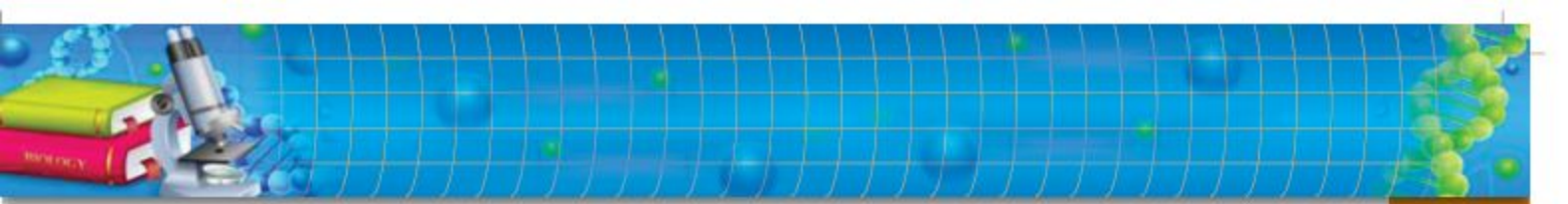
Epilepsy:

It is a brain disorder in which there is temporary alteration in one or more function or recurrent seizures. It is due to the abnormal electrical activity in brain. Stimulus like sudden flash light on eyes is also associated cause of the epilepsy. As a result, the victim may become unconscious with stiffness of the body and then twitches or jerks uncontrollably. The frequency of seizures can be reduced by using proper medication.



SUMMARY

- ◆ The changes or stimuli are detected by special cells or organs termed as **RECEPTORS**.
- ◆ The activity or response performed after analyzing stimulus is exhibited by organs like muscles, glands which are termed as effectors.
- ◆ There are two types of coordination:
1) Chemical Coordination, 2) Nervous Coordination
- ◆ The neurons upon stimulation generate electrochemical signal.
- ◆ Lower organisms and plants have chemical coordination through signaling molecules.
- ◆ Human nervous system like other vertebrates is “centralized-type nervous system”
- ◆ Brain is the major command and control center of our body.
- ◆ Human brain consists of following important parts: Cerebrum, cerebellum, thalamus, hypothalamus and medulla oblongata.
- ◆ Spinal cord is a butterfly-shaped, thick, whitish, long tube like structure which arises from medulla oblongata and extends down through the vertebral column.
- ◆ Peripheral Nervous System consists of cables or Nerves which arise from the central nervous system and connect it to different organs of the body.
- ◆ The cells of the nervous system are termed as neurons.
- ◆ Each neuron is specialized to generate and conduct neuronal signal or nerve impulse.
- ◆ The pathway of a reflex action is termed as reflex arc.
- ◆ Human body can detect number of stimuli like light, sound, gravity, damage to the tissues, etc, through its various receptors.
- ◆ Accommodation is an automatic process of altering focus to get sharper image of the near objects.
- ◆ Vitamin A or retinal is required for proper vision.
- ◆ Short-sightedness (myopia) refers to the difficulty in focusing distant object while the near objects are focused normally. On the other hand long-sightedness (Hyperopia) is the difficulty in focusing closer objects.

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- ◆ Colour blindness is a deficiency of vision in which one cannot distinguish certain colors such as blue and yellow or red and green.
 - ◆ Ear is an organ of hearing and balance.
 - ◆ Endocrine system consists of ductless glands which secrete hormones directly into the blood.
 - ◆ Pituitary gland located in brain secretes number of hormones which influence upon other endocrine glands also besides other organs.
 - ◆ Thyroid gland, a butterfly shaped gland located on trachea in the base of neck secretes thyroxine and calcitonin.
 - ◆ Pancreas consists of patches of cells called "Islet's of Langerhans" which are involved in regulating glucose metabolism through their hormones insulin and glucagon.
 - ◆ Diabetes mellitus is a disorder in which pancreas produces insufficient or no insulin.
 - ◆ Testes secrete testosterone which is responsible for the development of secondary characteristics in boys.
 - ◆ Estrogen is responsible for the development of secondary characteristics in girls, etc. Progesterone maintains and prepares uterus for pregnancy.
 - ◆ Hormone secretion is regulated through feed-back control so that they are secreted whenever required.
 - ◆ Paralysis is partial or complete loss of controlled movement caused by the inability to contract one or more muscles.
 - ◆ Epilepsy is a brain disorder in which there is temporary alteration in one or more function or recurrent seizures.

EXERCISE

A. MULTIPLE CHOICE QUESTIONS

Choose the correct answer:

- i) The activity in relation to changes in environment is:
(a) Stimulus (b) Response
(c) Both a & b (d) None of these
- ii) Stimulus is detected by:
(a) Receptor (b) Effector
(c) Nerve (d) All of these
- iii) The type of coordination through electrochemical signals is:
(a) Nervous (b) Chemical
(c) Mechanical (d) All of these
- iv) The chemicals released from one cell and carried to signal some distant cell through blood are:
(a) Neurotransmitters (b) Enzymes
(c) Hormones (d) all of these
- v) The type of coordination exhibited by Plants:
(a) Nervous coordination (b) Chemical coordination
(c) Mechanical coordination (d) Both a & b
- vi) The part of brain involved in reasoning is:
(a) Fore brain (b) Cerebrum
(c) Cortex (d) Frontal lobe
- vii) The part of brain involved in balance and precision in movements is:
(a) Cerebrum (b) Cerebellum
(c) Thalamus (d) Medulla oblongata
- viii) Vital functions for survival of animals are regulated by:
(a) CNS (b) PNS
(c) somatic sub-division (d) Autonomic sub-division
- ix) The shortest path of reflex action consists of:
(a) 1 neuron (b) 2 neurons
(c) 3 neurons (d) many neurons
- x) The type of lens in our eye is:
(a) Convex (b) concave
(c) Both a & b (d) None of these
- xi) The automatic process of altering focus to get sharper image of near object is:
(a) Vision (b) Accommodation
(c) Focus (d) All of these

- xii)** The vitamin necessary for proper vision is:
(a) Vitamin A (b) Vitamin B
(c) Vitamin C (d) Vitamin D
- xiii)** A colour blind person cannot see:
(a) Anything (b) Red
(c) White (d) Black
- xiv)** The book "Kitab-ul-manazir" was written by:
(a) Jabir bin Hayan (b) Ibn-al-Haitham
(c) Ali Ibn-Isa (d) Bu-Ali Sina
- xv)** Sensory hair-cells are present in:
(a) Retina (b) Cochlea
(c) Skin (d) Nose
- xvi)** The gonads are the target organ for:
(a) FSH (b) LH
(c) Both a & b (d) None of these

B. SHORT QUESTIONS:

- i)** What do you mean by feedback system?
- ii)** Why the nervous coordination is faster than chemical coordination?
- iii)** Which of the two coordination types is better and why?
- iv)** How reflex action works by a reflex-arc?
- v)** Why driving license is not issued to a color blind person?
- vi)** Why thyroid gland swell up and give the name of the disease?
- vii)** What is the role of Islet's of Langerhans cells?
- viii)** What is "emergency hormone" and why it is named so?
- ix)** What are the possible reasons of paralysis?
- x)** What is Epilepsy?
- xi)** Differentiate between Chemical and Nervous Coordination.

C. EXTENSIVE RESPONSE QUESTIONS:

- i)** Discuss the detailed structure of human brain?
- ii)** Write a note on human eye or ear.
- iii)** Why does the intelligence level of persons differ from each other, even though the components of brain are same?
- iv)** What is endocrine system? Discuss the gland involved in regulation of blood glucose and how?
- v)** What is diabetes mellitus? Explain some ways of its management.

Chapter 4

SUPPORT AND MOVEMENT

Major Concept

In this Unit you will learn:

- Introduction
- Concept of Movement
- Concept of Support and its Requirement
- Types of Skeletons
- Human Skeleton (Axial Skeleton and Appendicular Skeleton)
- Ball-n- Socket and Hinge Joints
- Action of Antagonistic Muscles at a Hinge (Elbow) Joint
- Disorders of Skeletal System (Arthritis and Osteoporosis)

